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Group Report

MakeMe

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11. Project Summary and Motivation

Problem
Maintaining a healthy lifestyle requires consistent effort, including regular physical activity, adequate hydration, and balanced nutrition. Traditional fitness applications provide users with tools to monitor their activities, record their meals, and track their progress, but users may often struggle to remain motivated to use these tools over time.

Proposed Software
MakeMe is a gamified fitness application that extends conventional fitness-tracking functionality into life-simulation mechanics and exaggerated fantasy character progression. Its defining feature is a customizable avatar whose daily and long-term appearance responds to the users recorded health related behaviors. Which draws on the concepts of vicarious reinforcement and identification through virtual self-modeling to visually reinforce healthy behaviors and encourages users to identify with their virtual representation [1].

Intended Users
MakeMe is primarily intended for individuals who enjoy video games and want to establish healthier daily habits through an interactive and rewarding experience. Potential users are limitless though it is mainly targeted to individuals who find conventional fitness applications mundane or difficult to maintain.

Value of the Project
The primary value lies in combining health monitoring with a persistent character progression system, adding a motivational layer by transforming general fitness application features into visible fantasy based character development.

Motivation
My motivation for pursuing this project comes from my interest in video game development and my experience with conventional fitness applications, where tracking numerical data alone provided limited satisfaction. This led me to consider how game design principles, particularly character progression could serve as visual rewards, which would make health monitoring more engaging.

12. Background and Related Work

MakeMe's avatar system draws on virtual self-modeling through vicarious reinforcement by Fox and Bailenson, which explored how observing virtual representation of oneself receiving reinforcement would influence exercise behavior [1]. However, their research used virtual representation that resembled the participants, whereas MakeMe features fantasy-based character development that would later skew away from the users physical appearance. This distinction is intentional, as MakeMe allows users to watch their avatars develop into increasingly muscular and powerful characters, potentially inspiring them to aspire to the strength and appearance of their virtual counterparts, motivating them to maintain healthy habits. This design aligns the concept of wishful identification mentioned in the review research by Praetorius and Gorlich on the Proteus Effect, which describes how an avatar's characteristics may influence a users behavior and self-perception [2]. While existing research does not directly establish a link between avatar progression and the development of healthy habits, incorporating them may still offer motivational benefits compared with health tracking applications without an avatar.

What's Out There

Although ranking of the top 10 fitness trackers may vary due to subjective reviews and potential biases, many of them can be broadly classified into three categories based on their approaches to fitness tracking and gamification.

Category 1 consists of conventional fitness trackers that incorporate little to no gamification, focusing primarily on workouts, activity logging, and social features [3], [4]. Their popularity suggests that stand-alone fitness monitoring remains an established approach to helping users track their progress and maintain their fitness routine. However, this category represents only one approach to fitness tracking and does not establish whether users would benefit from engaging or game-oriented alternatives.

Category 2 consists of challenge-based gamified fitness trackers that are oriented to encourage physical activity through step-count goals, competitions, social challenges, and real-world incentives. In 2020, (Neupane et al.) reviewed 103 gamified fitness tracker apps and found that challenges, competition and social influence frequently appeared together, while social features played a central role across nearly every identified game genre [5]. These findings demonstrate how conventional activity tracking can be enhanced with game mechanics that encourage users to remain active through external rewards, social interaction, and competition. However, this approach primarily builds on existing fitness tracking activities rather than making exercise part of a broader, evolving game experience.

Category 3 consists of stand-alone games that integrate physical activity directly into gameplay, encouraging users to exercise as they pursue in-game objectives. Games such as PokemonGO and Wii Sports demonstrate the popularity of this approach making physical activity more engaging through exploration and interactive gameplay [6], [7]. However, research on PokemonGO suggests that increased in physical activity may be short-lived, highlighting the challenge of sustaining user engagement and long-term exercise habits.

MakeMe aims to combine elements from all three categories into a unified experience. It aims to incorporate the vital features of health and activity monitoring from conventional fitness trackers, the measurable goals and rewards of challenge-based gamification, and the immersive progression of stand-alone games. Its defining feature of an ever-evolving, customizable fantasy avatar that becomes increasingly muscular and powerful as users maintain healthy habits. Rather than relying solely on activity statistics, social influence, or individual game missions, MakeMe proposes to investigate whether aspirational character development can provide an additional source of motivation for users to maintain their health routines over time.

13. Implementation Goals and Scope

13.1. Implementation Goals

1. **Goal 1.** Implement an independent body-part-based avatar progression. A customizable avatar whose initial build is determined by a user survey with individual body parts grows independently in response to logged workouts. Workout duration, selected muscle groups, and some aspects in meal logging influences avatar growth, while a decay system will regulate its development overtime.

2. **Goal 2.** Integrate core functionality from conventional fitness tracker apps and Avatar state systems. Implementing essential activity logging for habit tracking such as workouts, meals, hydration, and sleep logging systems that influence the avatar's physical condition and appearance. The system will incorporate workout streaks, clean, and dirty bulk progression,

and conditional states such ass sleepy, dehydrated, in peak condition, recovering and dead.

3. **Goal 3.** Develop and a three-month progression cycle and reward system. Implement a 90-day progression period in which users can track their habits and observe changes in their avatars. At the end of the cycle, the system will present a summary of the user's progress and determine whether specific user goals were accomplished for a new cycle in which an ascension system activates and resets the avatar through another user survey. Limited badges and equipment are rewarded to users for completing daily and long-term goals through a reward system.

13.2. Core Functional Requirements

- **R1.** Initial Avatar Set Up. The system will collect initial user data through a survey and use the response as the avatars baseline.
- **R2.** Workout Logging and Muscle-Specific Growth. A dedicated workout logger that collects duration, selected exercised body parts, and intensity level from user input. The system uses this input to calculate growth for selected body parts.
- **R3.** Workout Streak Tracking: The system maintains a workout streak using a 48-interval after a workout log, allowing users to include rest days without automatically breaking their streaks.
- **R4.** Avatar Decay. The avatar decays continuously and can be accelerated through poor meal logging habits. This system balances growth from workout logging to simulate a semi-realistic growth rate.
- **R5.** Avatar State System: The system updates the avatars' appearance and state based on the user's logged habits.
- **R6.** Habit Logging. Allows users to record meals, hydration and sleep that contributes to avatars state system and progression.
- **R7.** Bulk Progression. Accumulated throughout the cycle and reads through macro report from macro tracker and calculates progression in clean bulk and dirty bulk that influence the avatars overall appearance.
- **R8.** Recovery and Revival. Prolonged period of inactivity can result to the avatars death. A recovery system detects the users return within three stages that determines recovery state from initial return work out logging to a three-day workout streak. The avatar loses all its progress and will return to its initial state.
- **R9.** Cycle Completion and Summary. The system tracks a 90-day cycle to simulate a three-month transformation and presents a summary of the user's activity, avatar development, and achievements.
- **R10.** Ascension System. Determines if the user has completed their long-term goals and allows the user to begin a new cycle, including a new initial survey and avatar baseline. A badge or title will be rewarded with each ascension.
- **R11.** Limited rewards such as badges or cosmetic items for completing certain milestones.

13.3. Initial Scope

In scope. The initial version of MakeMe will focus on implementing the avatar's initial configuration, muscle-specific growth, workout logging, streak system, and time-based decay.

It will also include habit logging, supporting avatars conditional states and bulk progression. A 90-day progression cycle, cycle-completion and summary that provides the user a long term objective.

Out of scope. The initial implementation will exclude extensive equipment collections, multi-player features, competitive leaderboards, and advanced social systems. Character customization, avatar appearance states, and visual progression will also be limited to a predefined set of options. Additionally, the application will not provide medical diagnoses or claim that avatar development accurately represents a user's real body composition or physical health.

14. Initial Design

14.1. Use-Case Diagram

The primary actor is the User, who interacts with the application to track health habits and monitor personal progress. The user's goals include creating an initial avatar, logging health related-activities, viewing the avatars visual development as it changes in stature, appearance, and condition in response to their habits, reviewing macronutrient summaries and cycle progress, avatar personalization, and managing health goals. These use cases illustrate how MakeMe combines health habit tracking with visual character progression to encourage users to maintain healthy routeines throughout the 90-day progression cycle represented in figure 1.

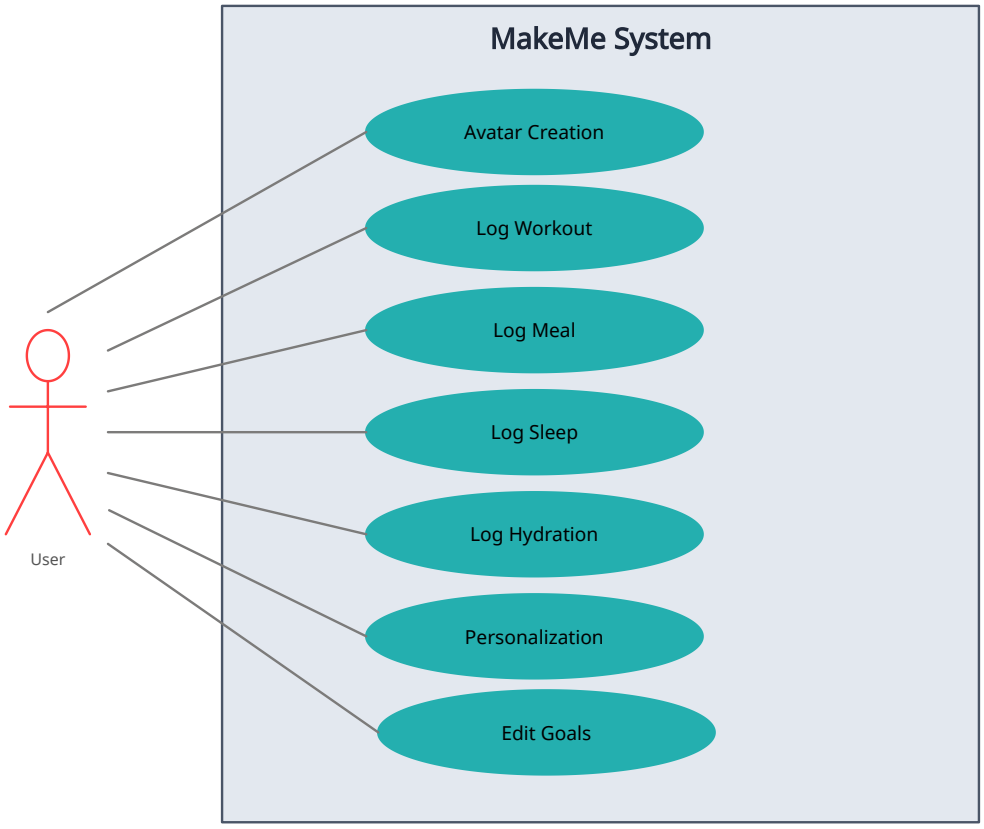


Figure 1: Initial use-case diagram for the proposed system.

14.2. Class Diagram

The class diagram shown in figure 2 organizes MakeMe into classes responsible for user profiles, avatar progression, workout logging, nutrition tracking, and health habits. The Avatar and BodyPart classes manage visual development, while BulkProgression and the streak classes track consistency and progress. The CycleManager manages the 90-day progression cycle, and the RewardSystem handles milestone rewards and equipment unlocks. Relationships between classes primarily use associations and composition, while inheritance connects the abstract Streak class to WorkoutStreak and MealStreak. Enumerations define predefined states and options used throughout the system.

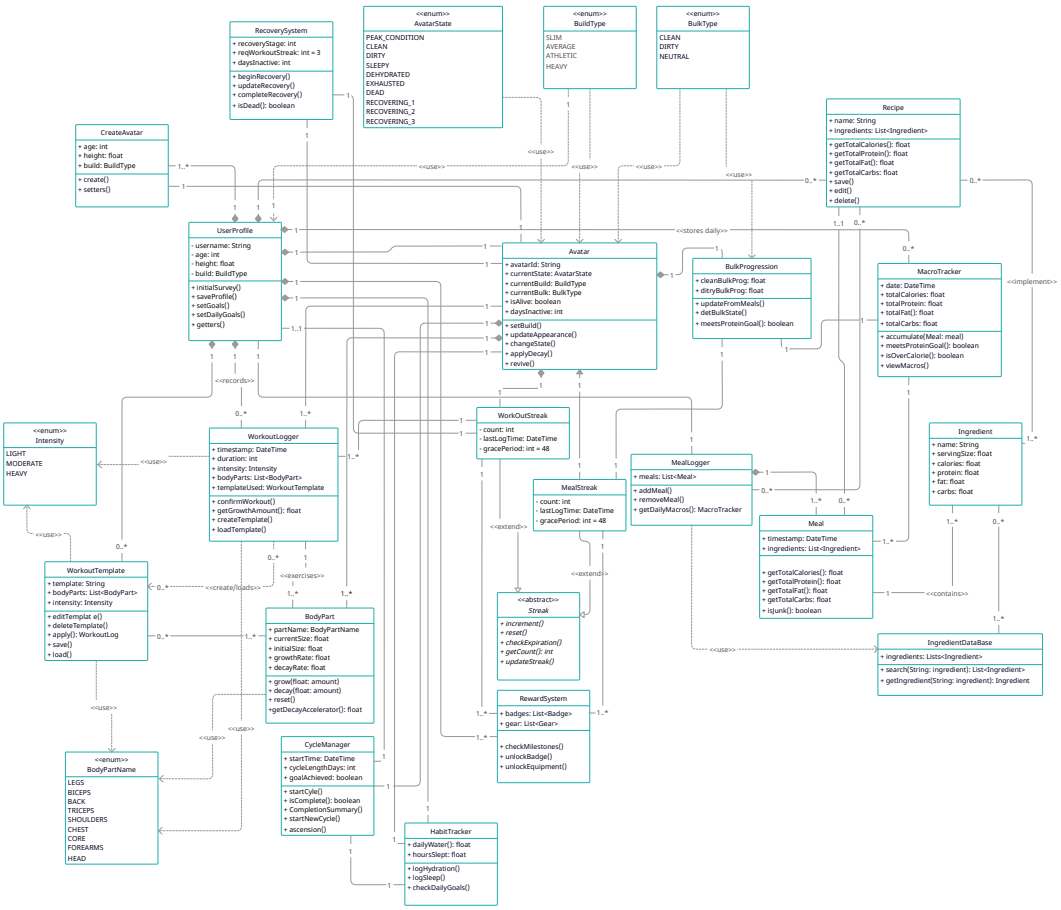


Figure 2: Initial UML class diagram for the proposed system.

15. Feasibility and Supporting Evidence

The technical feasibility of MakeMe depends primarily on developing its avatar progression system, which is expected to require the largest portion of development time. Unlike conventional fitness applications that display numerical statistics, MakeMe aims to visually represent progress through body-part-specific growth. The system must translate workout duration, intensity, targeted muscle groups, consistency, and inactivity into visible changes to the avatar's chest, arms, shoulders, back, core, and legs. To reduce complexity, the initial implementation will use a predefined 2D or semi-2D character model with limited growth stages for each body part, avoiding complex real-time 3D deformation. The project will also exclude multiplayer features, competitive leaderboards, extensive equipment collections, advanced social systems, and unrestricted character customization. Additionally, the application will not provide medical diagnoses or claim that avatar development accurately represents a user's real body composition or physical health. These exclusions allow development to focus on the core features of habit tracking, avatar progression, and the 90-day cycle. An AI generated UI mockup by ChatGPT can be used to visualize the interface and avatar progression, providing a reference for development and user testing as shown in Figure figure 3.

For nutrition tracking, MakeMe could integrate an external API such as USDA FoodData Central[8] to retrieve food information and calculate calories and macronutrients, reducing the need to create a comprehensive food database from scratch. However, the team must account for API integration, serving-size matching, and secure API key management; a smaller predefined database could serve as a fallback. Flutter is a potential framework for developing both Android

and iOS applications from a shared codebase, although learning Dart and Flutter introduces an initial learning curve. Previous experience with Java should help because both languages support object-oriented programming, classes, methods, and familiar control structures. Development can initially focus on Android using Android Studio's emulator to test the interface and core functionality without requiring a physical device. However, iOS development and testing will still require access to the appropriate Apple development tools. Overall, MakeMe is feasible if the team prioritizes a functional minimum viable product, tests the avatar growth and decay rules carefully, and postpones nonessential features until the core tracking and progression systems work reliably.



Figure 3: Supporting feasibility evidence for the proposed project.

16. Resources and Risks

Table 1: Initial project risks and mitigation strategies.

Risk / Constraint	Why it matters	Mitigation / Fallback
Replace with risk 1	Explain the impact	Explain how the risk could be reduced or handled.
Replace with risk 2	Explain the impact	Explain how the risk could be reduced or handled.

References

[1] J. Fox and J. Bailenson, "Virtual self-modeling: The effects of vicarious reinforcement and identification on exercise behaviors," *Media Psychology - MEDIA PSYCHOL*, vol. 12, pp. 1–25, Mar. 2009. DOI: 10.1080/15213260802669474

[2] A. S. Praetorius and D. Görlich, "How avatars influence user behavior: A review on the proteus effect in virtual environments and video games," in *Proceedings of the 15th international conference on the foundations of digital games*, 2020, pp. 1–9.

[3] A. Hall. "The 10 best workout and fitness apps of 2026," Forbes Health, Accessed: Oct. 12, 2026. [Online]. Available: https://www.forbes.com/health/weight-loss/best-fitness-apps/

[4] H. Herman. "The 9 best fitness apps in 2026," Zapier, Accessed: Oct. 12, 2026. [Online]. Available: https://zapier.com/blog/best-fitness-tracking-apps/

[5] A. Neupane, D. Hansen, A. Sharma, J. A. Fails, B. Neupane, and J. Beutler, "A review of gamified fitness tracker apps and future directions," in *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*, 2020, pp. 522–533.

[6] G. K. Koh, K. M. Y. Low, S. H. Ali, H. S. Cheng, and I. Y. H. Ang, "Motivational factors, physical activity impacts, and sociopsychological effects of pokémon go in players over the years: Scoping review," *Journal of Medical Internet Research*, vol. 28, no. 1, e89235, 2026.

[7] J. Yim and T. N. Graham, "Using games to increase exercise motivation," in *Proceedings of the 2007 conference on Future Play*, 2007, pp. 166–173.

[8] U.S. Department of Agriculture, Agricultural Research Service, *FoodData Central API Guide*, https://fdc.nal.usda.gov/api-guide, Accessed: 14 October 2026.